

EcoInnovators – Solar Panel Detection Model

Project Name: EcoInnovators – Solar Panel Detection Model

Event: Ideathon 2026

Team: Void Crew

GitHub Repository: https://github.com/Premithius/EcoInnovators_Ideathon_2026_Void_Crew

Model Type: YOLOv8s-seg (Instance Segmentation)

Application Domain: Sustainability, Renewable Energy, Smart Cities

1. Model Overview

The EcoInnovators – Solar Panel Detection Model is a computer vision system designed to automatically detect and segment rooftop solar panels from aerial and satellite imagery. The model supports scalable assessment of solar energy adoption by providing accurate, location-aware identification of rooftop-mounted photovoltaic installations across large geographic areas.

Manual surveying of solar infrastructure is costly and often infeasible at city or regional scales. This model addresses the challenge by leveraging deep learning–based instance segmentation to identify solar panels with pixel-level precision. The resulting outputs enable downstream analysis such as rooftop solar coverage estimation, urban energy planning, and renewable infrastructure monitoring. The project was developed by Team Void Crew as part of Ideathon 2026, with a focus on practical deployment and sustainability-driven impact.

2. Intended Use and Users

The model is intended for infrastructure-level analysis of rooftop solar installations using overhead imagery. It is suitable for urban, suburban, and semi-urban environments where rooftop solar panels are visible in aerial or satellite data.

Intended users include urban planners, renewable energy analysts, sustainability researchers, smart city developers, and governmental or institutional bodies involved in energy planning. The model may also support academic research and clean energy startups focused on solar adoption mapping and environmental impact assessment.

The system is not designed for individual household surveillance, personal data analysis, or enforcement-related applications. All use cases should focus on aggregated insights rather than

property-level decision-making.

3. Model Architecture and Design Choices

The model is built on YOLOv8s-seg, a single-stage instance segmentation architecture optimized for a balance between accuracy and computational efficiency. The “small” variant was selected to enable faster training and inference while maintaining reliable segmentation performance.

YOLOv8s-seg integrates object detection and mask prediction within a unified framework, allowing simultaneous generation of bounding boxes and segmentation masks. This design is well suited for rooftop solar panels, which vary in size, orientation, and layout. An input resolution of 640×640 pixels was chosen to preserve rooftop-level detail while maintaining manageable memory requirements. Large satellite images are processed through tiling to ensure consistent inference over wide areas.

4. Data Description and Preprocessing

The dataset consists of aerial and satellite imagery containing rooftops with visible solar panel installations. Images include diversity in building structures, roof materials, panel orientations, and environmental conditions such as lighting and shadows to improve generalization.

Annotations are provided as instance-level segmentation masks, with each rooftop solar installation labeled individually. Only rooftop-mounted panels are included to maintain task specificity. Large images are divided into 640×640 tiles prior to training. Preprocessing includes pixel normalization, annotation consistency checks, and data augmentation techniques such as random flipping, scaling, and brightness variation to enhance robustness.

5. Training Procedure and Environment

Training is conducted using the Ultralytics YOLOv8 framework built on PyTorch. The model is initialized with pretrained weights to leverage transfer learning and improve convergence speed.

Training is performed in a GPU-enabled environment with mixed-precision support to optimize computational efficiency. Hyperparameters are selected based on training stability and validation behavior rather than aggressive optimization. The training strategy prioritizes reproducibility, robustness, and practical deployment considerations.

6. Evaluation Metrics and Performance Characteristics

Model evaluation follows standard object detection and segmentation practices, including Intersection over Union (IoU) and mean Average Precision (mAP), assessed on a held-out validation subset.

Given the ideathon context, evaluation emphasizes qualitative and practical performance. The model reliably detects and segments rooftop solar panels in clear imagery, producing masks that align closely with panel boundaries. Performance degrades under challenging conditions such as heavy shadowing, extreme roof angles, or low-resolution imagery, consistent with known limitations of aerial image analysis.

7. Limitations and Failure Cases

The model may misclassify rooftop elements that visually resemble solar panels, such as skylights, dark roof tiles, or reflective equipment. Occlusions from vegetation, nearby buildings, or strong shadows can result in incomplete or missed detections.

The system is not designed to detect ground-mounted solar farms or non-rooftop installations. Performance may also vary across regions with rooftop styles not represented in the training data. Predictions should be validated before use in high-impact planning or policy contexts.

8. Ethical, Privacy, and Safety Considerations

The model operates exclusively on overhead imagery and does not process personal identifiers or biometric data. Nevertheless, rooftop imagery can still be sensitive, and users must comply with local regulations and satellite data usage policies.

The system should not be used for surveillance or targeted monitoring of individual properties. Its intended purpose is aggregated infrastructure analysis supporting sustainability and renewable energy objectives.

9. Environmental Impact

The project contributes positively to environmental sustainability by enabling scalable analysis of rooftop solar adoption. Insights derived from the model can inform renewable energy expansion, carbon reduction strategies, and smart city planning.

From a computational perspective, the use of a lightweight YOLOv8s architecture reduces energy consumption during training and inference compared to larger models. Training was conducted efficiently and within reasonable computational limits.

10. Usage Instructions

Users can clone the GitHub repository and install dependencies as documented. Input imagery should be provided as 640×640 tiles or larger images suitable for automatic tiling during inference.

The model outputs bounding boxes and segmentation masks in YOLOv8 format, which can be visualized directly or exported for GIS-based spatial analysis. Fine-tuning on region-specific datasets is recommended for improved performance.

11. License and Attribution

This project is distributed under the license specified in the GitHub repository. Users must comply with the license terms and provide appropriate attribution to Team Void Crew and the EcoInnovators project.

All third-party libraries and frameworks, including YOLOv8, are used in accordance with their respective open-source licenses.

12. Future Work and Improvements

Future work includes expanding the dataset to additional geographic regions, improving robustness under challenging lighting conditions, and integrating rooftop area estimation for solar capacity analysis. Planned enhancements also include tighter GIS integration and deployment as an API or analytics service for smart city and sustainability applications.